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TERTIARY GEOMAGNETIC FIELD REVERSAL IN SCOTLAND

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Twenty-one examples in which the magnetic polarities of baked contact rocks (sediments, conglomerates and dykes) agree with those of the igneous bodies which baked them (Scottish Tertiary dykes) are presented. These agreements strengthen the evidence in favour of field reversal. If field reversal has operated solely or predominantly, almost all baking-baked pairs should agree in polarity since each component of each pair presumably cooled in the same ambient field. If self-reversal has mainly operated, however, there should be a large number of polarity disagreements between the baking and baked rocks since rocks of different character would not necessarily both have self-reversed.

The Tertiary pole position derived from the paleomagnetic data from both baking and baked rocks was 73.4°N, 196.8°E.

1. INTRODUCTION

It is well-known that approximately 50% of the world's rocks are magnetized normally, that is, in directions consistent with the earth's dipole moment in its present sense, and that the other 50% are reversely magnetized, in directions consistent with a reversed dipole moment of the earth. There are apparently only two possible explanations of the reversed magnetizations, namely either (i) that for approximately 50% of geological time the earth's dipole moment has actually been reversed (field reversal), or (ii) that the reversely magnetized specimens possess, or have possessed, some intrinsic property which has either enabled them to become magnetized in directions opposite to those of the ambient fields in which they were laid down, or has enabled normal magnetizations to reverse at some time subsequent to their acquisition (self-reversal) [1, 2].

The evidence for and against field reversal and self-reversal has been reviewed by Wilson [3, 4]. In favour of the predominance of field reversal is the evidence from baked contacts [3-5], reversals "caught in the act" [6-13] and the dating of polarity epochs [9, 14-23]. On the other hand petrological differences between normal and reversed specimens [24-30] suggest the possibility of self-reversal. Few systematic differences in the magnetic properties of normal and reversed specimens have yet been discovered, apart from the directions of the NRMs. A paradox not yet resolved is that certain rock sequences which exhibit petrological differences between normal and reversed specimens (suggesting self-reversal) also produce independent evidence of field reversal [4, 30].

2. GEOLOGY AND COLLECTION DETAILS

The rocks utilised in this study were from the Great Cumbrae Island (Buteshire) and the Ayrshire and Renfrewshire coasts (56°N, 5°W). The area comprises mainly Old Red Sandstone and conglomerate of Upper Old Red Sandstone age into which dykes, sills and vents intrude [31-34]. There are, in particular, several generations of dykes, namely, (i) porphyritic and non-porphyritic basalt, bostonite and felsite dykes of Calciferous Sandstone age which strike between approximately ENE and NE, (ii) quartz-dolerite Permo-Carboniferous dykes striking approximately E-W, and (iii) olivine dolerite, crinanite, andesitic-basalt and tholeiitic-basalt dykes of Tertiary age which strike between approximately N and NW.

Only rocks of Tertiary age were collected for this work; and in order to exclude older rocks, only those dykes striking between 300 °ETN and 360°ETN were sampled (with one exception described below). Two oriented lump samples were collected from each of 17 different dykes, together with two oriented lump samples of baked material from within 6 cm of the edge of each dyke. Contact samples were taken from this close to the dyke edges to ensure that only

Table 1
Locations of Scottish Tertiary dykes and baked contacts.

System	*Location*
A	Cloch Point, Renfrewshire.
C-F	Skelmorlie, Ayrshire.
G	Inverkip, Renfrewshire.
K	Largs, Ayrshire.
L-W, Y, Z	Great Cumbrae Island, Buteshire.

the most highly baked material was collected. Frequently, contact red sediments were bleached to a buff colour up to several centimetres from the dyke edges: and this type of material was sampled wherever possible.

A four-dyke composite system at Skelmorlie (Ayrshire) was also sampled. A narrow dyke C (tapering from 40 cm to 5 cm in width) striking approximately NE cut three NW trending dykes D, E and F (each 90 cm wide). Potassium-argon dating (see below) has confirmed that dyke D is Tertiary (as expected from its NW strike direction) and has indicated that dyke C is also Tertiary (despite its NE strike direction). Samples from all four dykes and their baked contacts have been used here.

In all, four distinct types of baked material were included in the collection, namely (i) Old Red Sandstone, (ii) conglomerate of Upper Old Red Sandstone age, (iii) very fine-grained grey sediment, and (iv) basalt dyke. The locations of the dyke-contact systems are given in table 1.

3. AGE CONTROL

Eight of the dykes (C, D, K, N, S, U, V, Y) have been dated by Dr. R. L. Grasty of Cambridge University using the potassium-argon technique [35]. Mean ages lay in the range 33.9 m.y. to 57.1 m.y. (table 2) thus confirming the early Tertiary intrusion.

4. MEASUREMENTS

Two oriented core specimens (1 in. long by $\frac{7}{8}$ in. diameter) were cut from each lump sample. One specimen from each dyke and each contact, selected at random from those available, has been thermally demagnetized; and two further specimens from each dyke, selected from those remaining, have been ac demagnetized. In addition, a few other specimens which have been thermally demagnetized in connection with the determination of ancient field intensities [36] have been included in the analysis of results.

Table 2
Mean potassium-argon ages of dyke specimens. Errors are standard deviations and include errors in determination of potassium and the volume of radiogenic ^{40}Ar.

Dyke	Mean age (m.y.)
C	44.3 ± 3.4
D	51.4 ± 3.5
K	53.3 ± 1.7
N	51.0 ± 6.2
S	40.9 ± 2.6
U	42.4 ± 4.1
V	33.9 ± 1.2
Y	57.1 ± 1.1

Twenty-five dyke specimens and 32 baked contact specimens were thermally demagnetized in zero field using a furnace in conjunction with an astatic magnetometer. Vector heating diagrams were plotted at 50°C temperature intervals; and in each case the initial NRM direction was obtained from the high temperature regions of the curves, above any instability. In all but three dykes (L, N and R) stable directions were reached by 270°C; and in all baked contacts stable directions were reached by 100°C. Initial and final directions of magnetization are given in table 3 (dykes) and table 4 (baked contacts). Baked contacts from O, R and W were too weakly magnetized to enable heating curves to be produced. In these cases only the uncleaned directions are presented.

Forty-two dyke specimens were ac demagnetized. The optimum demagnetizing field was determined by progressively demagnetizing five specimens at 50-oersted peak field intervals from zero to 250 oersteds and calculating the Fisher [37] precision parameter estimate (k) for all five specimens at each field. k was found to be greatest (indicating smallest scatter) at 150 oersteds peak field. The remainder of the specimens was then measured at 0 and 150 oersteds. Initial and final (150-oersted) directions of magnetization of the dykes are given in table 3.

5. DISCUSSION

Of the 21 dykes investigated, 20 were found to be reversely magnetized and only one normally magnetized (dyke N) after cleaning. This proportion is consistent with data published hitherto which indicate that over 90% of British Tertiary rocks are reversed [38].

In all cases the polarities of the baked rocks agreed with those of the corresponding baking dykes. In table 4 the absolute angles (θ) between

Table 3
Directions of magnetization of Scottish Tertiary dykes before and after cleaning. Positive inclinations are below horizontal (normal); negative inclinations are above horizontal (reversed).

Specimen	*Before demagnetization*		*After thermal demagnetization*		*After ac demagnetization*	
	declination (°ETN)	inclination (deg)	declination (°ETN)	inclination (deg)	declination (°ETN)	inclination (deg)
IA2	177.5	-61.0			176.5	-63.5
IIA1	171.0	-53.5			169.5	-59.0
IIA2	180.0	-55.0	183.0	-60.0		
IC2	181.5	-51.0	185.5	-63.0		
IIC2	239.5	-57.0			217.0	-66.0
VC2	181.5	-75.5			182.0	-76.5
IID2	149.0	-11.0			144.5	-30.5
IIID1	146.0	-13.5			146.5	-36.0
IIID2	146.0	-24.5	166.5	-29.0		
IE1	185.0	-44.0			189.0	-52.5
IE2	187.5	-47.0	186.0	-56.0		
IIE2	197.5	+45.0			186.0	-52.0
IF1	196.0	+71.0			186.5	-65.0
IF2	133.0	-32.5	183.5	-62.0		
IIF2	165.5	+64.0			177.5	-56.0
IG2	131.0	-45.0			134.5	-61.5
IIG1	163.5	+61.5			175.5	-61.5
IIIG2	159.0	-71.5	173.0	-68.0		
IK1	168.5	-63.5			167.0	-65.5
IK2	157.5	-68.5	154.5	-78.0		
IIK1	180.5	-33.0			174.5	-48.0
IIL1	353.5	-21.5			138.0	-58.0
IIL2	13.0	-27.0	137.5	-60.5		
IVL2	16.0	+22.0			134.5	-59.0
IM1	133.0	+47.5			105.5	-39.0
IM2	71.5	+47.0	94.5	-57.5		
IIM1	110.0	+65.5			236.0	-60.5
IN1	182.5	+88.0			324.5	+71.5
IN2	280.0	+82.0	343.0	+55.0		
IVN1	356.0	+82.0			332.5	+58.0
IVN2	113.0	+13.5	296.0	+22.0		
IO1	227.5	+35.0			205.5	-52.5
IO2	209.0	+66.5	233.0	-63.0		
IIO2	191.5	+61.0			159.0	-59.5
IP1	121.0	+81.5			219.0	-54.0
IP2	218.5	+64.5			229.0	-51.0
IIIP1	353.5	-77.5	159.5	-66.5		
IIIP2	251.0	-68.5	164.0	-59.0		
IIQ1	149.0	+37.0			160.0	-42.5
IIQ2	158.0	-14.5	157.0	-47.0		
IVQ2	164.5	-48.5			158.0	-55.0
IR2	164.5	+86.0			197.0	-74.5
IIR1	175.0	+82.5			219.5	-63.0
IVR2	190.0	+71.0	219.0	-48.0		
IS1	116.0	-27.5			206.0	-78.0
IS2	322.0	-76.0	203.0	-74.5		
IIIS2	212.0	-64.0			225.5	-67.0
IIT1	336.5	+82.0			165.0	-81.0
IIT2	323.0	+78.5	241.0	-61.0		
IVT2	199.5	+68.5			210.5	-47.0
IU1	162.5	-52.5	165.0	-53.0		
IU2	157.0	-57.5	154.0	-56.0		
IIU1	148.5	+14.0			157.5	-27.5

Table 3 (continued)

Specimen	*Before demagnetization*		*After thermal demagnetization*		*After ac demagnetization*	
	declination (°ETN)	inclination (deg)	declination (°ETN)	inclination (deg)	declination (°ETN)	inclination (deg)
VIU2	175.0	-26.0			164.5	-48.5
IIV2	156.5	-55.5			154.0	-63.5
IIIV1	136.0	-19.0			167.5	-53.5
IIIV2	148.5	-34.0	172.0	-61.0		
IW1	154.0	-18.5			154.0	-23.0
IW2	150.0	- 8.5			152.0	-18.0
IIW1	166.5	-28.0	168.5	-27.0		
IIW2	169.0	-30.0	167.5	-32.5		
IIY1	182.0	-41.5			183.0	-66.5
IIY2	180.0	-42.0			178.0	-65.0
IIIY1	168.0	-71.0	213.0	-70.5		
IZ1	186.5	-63.5			184.5	-38.5
IZ2	196.5	-52.0	150.5	-43.0		
IIZ2	170.0	-30.0			173.0	-27.0

the directions of the thermally cleaned contacts and the directions of the corresponding thermally cleaned dykes are listed. In the three cases (N, P, U) where two dyke specimens have been thermally cleaned the mean dyke direction has been used to compute θ. The mean value of θ is $13^{\circ} \pm 6^{\circ}$ (standard deviation), and no dyke-baked pairs agree directionally within less than 2°. Three pairs have angular separations greater than 26° (twice the mean value).

θ is distributed randomly in both declination and inclination. It is therefore unlikely that the angular separations arise through errors in the magnetic orientation technique (in which case θ would be solely in declination) or because the contacts had not been heated above the Curie points of their constituents (in which case θ would only be in inclination). It is just possible that both of these processes have occurred, although there are other reasons for believing that contacts were heated above their Curie points. There were no noticeable discontinuities in the contact heating curves and no shape differences in the heating and cooling curves, although these would have been difficult to detect.

A mean angular separation of 13° is too large to be classed as perfect agreement. Such a large separation could be caused by errors in measurement due to inhomogeneity in the core specimens. In the furnace-astatic magnetometer system used the Z component of magnetization (along the core axis) was measured with the core in four different aspects beneath the magnetometer, whereas the X and Y components (X, Y and Z are mutually orthogonal) could only be meas-

Table 4
Directions of magnetization of Scottish baked contacts before and after cleaning. Positive inclinations are below horizontal (normal); negative inclinations are above horizontal (reversed). θ = angle between direction of baked contact and corresponding thermally cleaned dyke. ORSS = Old Red Sandstone.

Specimen	Type	*Before demagnetization*		*After thermal demagnetization*		θ^o
		declination (°ETN)	inclination (deg)	declination (°ETN)	inclination (deg)	
IVA1	ORSS	203.5	-55.5	195.0	-59.5	12
VIC2	ORSS	182.5	-54.0	207.0	-68.0	10
ID1	basalt dyke	176.0	-64.5	182.5	-63.0	2
ID2	basalt dyke	175.0	-67.5	181.5	-75.0	12
IVD1	conglomerate	135.0	-29.0	146.0	-36.0	13
IIIE2	basalt dyke	199.0	-68.0	199.0	-72.0	10
IVE2	conglomerate	166.0	-61.0	164.0	-63.0	13
IIIF1	basalt dyke	201.5	-33.0	195.0	-68.0	7
VF1	ORSS	172.0	-51.5	178.0	-52.0	10
VF2	ORSS	182.0	-59.5	195.5	-61.5	12
IVG2	grey sediment	161.5	-27.0	162.0	-27.0	41
VG2	ORSS	157.5	-67.0	158.0	-65.5	7
IVK1	ORSS	165.0	-64.5	173.0	-67.0	13
IVK2	ORSS	164.0	-64.0	181.0	-66.5	13
IIIL1	ORSS	154.5	-53.0	159.0	-53.0	15
IIIM2	ORSS	223.5	-61.5	219.5	-71.0	45
IIN1	ORSS	320.5	+55.0	320.5	+63.0	24
IIN2	ORSS	325.0	+58.5	321.0	+54.5	15
IIIO1	ORSS	210.0	-60.5	-	-	-
IIP2	ORSS	164.5	-57.0	169.0	-60.5	4
IVP2	ORSS	178.0	-63.5	173.5	-62.0	6
IQ1	ORSS	160.5	-46.5	160.0	-51.0	5
IQ2	ORSS	160.5	-49.0	164.5	-48.0	13
IIIR1	ORSS	117.0	-78.5	-	-	-
IVS2	ORSS	191.5	-57.5	195.0	-59.0	16
VS1	ORSS	217.0	-71.5	218.5	-73.0	5
IT1	ORSS	227.0	-47.5	235.0	-48.0	5
IIIU2	ORSS	149.5	-47.0	154.5	-46.0	9
IVU2	ORSS	158.5	-43.0	163.0	-48.0	7
IVV1	ORSS	170.0	-59.0	174.5	-59.5	2
IVV2	ORSS	156.0	-55.0	162.5	-62.0	4
IIIW2	ORSS	158.0	-21.5	-	-	-
IVY1	grey sediment	175.0	-62.0	172.5	-62.5	18
IVY2	grey sediment	174.0	-61.5	174.0	-65.5	15
IVZ1	grey sediment	186.0	-42.0	192.5	-52.0	29

ured in two aspects each. Two different measurements each of X and Y may have been insufficient to mean out inhomogeneities. On the other hand the ac demagnetized dykes were measured under a different astatic magnetometer and by spinning the cores in an attempt to eliminate the effects of inhomogeneity. Even so the mean angular separation of directions from pairs of ac demagnetized specimens from 21 dykes was 16° ± 10° (standard deviation), that is, greater than the angular difference between thermally cleaned dyke and contact directions.

It seems likely, therefore, that the large angular separations are due to some instability in either the dykes or contacts, or both. The dykes are less stable generally than the contacts. Initial and final dyke directions are shown in fig. 1 and initial and final baked contact directions are shown in fig. 2. Examination of these diagrams indicates that before cleaning the dyke directions were much more scattered than those of the corresponding baked contacts. The mean angles between the initial and final directions are (i) 55° for the dykes, and (ii) 7° for the baked contacts. Even after cleaning the contact directions were more tightly grouped than the dyke directions. This is indicated by figs. 1 and 2 and the values of the Fisher [37] precision parameter estimates [k] given in table 5. Large angular separations between dyke and contact directions could thus be due to instability, although it is possible that there is some other, as yet undiscovered, explanation.

Nevertheless the polarity agreement between dyke and baked contact directions of magnetization is valid in all cases. The mean direction of the 32 thermally cleaned baked contacts differs from that of the 25 thermally cleaned dykes by only 4°, which is less than α_{95} in each case and hence not significant (table 5). There are thus 21 dyke-contact systems in which the baked and baking rocks are in polarity agreement. Further, since the heat-treated and ac demagnetized dykes possess mean directions differing by only 2°, both methods of magnetic cleaning are valid in this case.

The polarity agreement in 100% of the cases is evidence of field reversal. Had self-reversal been predominant, approximately 50% of the pairs should have disagreed in polarity since a statistical examination of all the rocks in the world indicates an equal proportion of normal and reversed polarities. There remains the possibility that both the reversed dykes and baked contacts have undergone self-reversal. It is unlikely that there exists a self-reversal mechanism common to all the four types of baked contact examined here and to the dykes themselves. It is just possible that during baking the contacts were impregnated by dyke fluids which were the carrier of a common self-reversal mechanism, but this is unlikely because there was no evidence of titanomagnetite in four baked contacts examined petrologically.

The mean direction of magnetization for all 99 cleaned specimens has declination 172.0°ETN and inclination -59.3° (reversed polarity). The mean virtual geomagnetic pole position then has

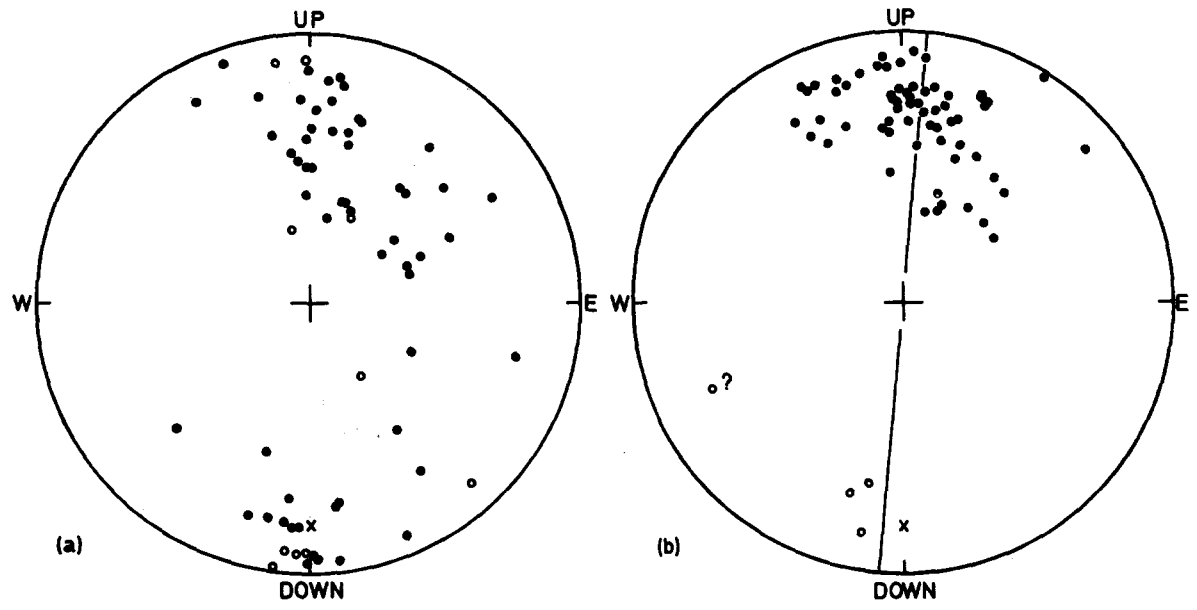


Fig. 1. Directions of magnetization of Scottish dykes (a) before cleaning, and (b) after cleaning. Equal area projections. Closed symbols represent southward declinations; open symbols represent northward declinations. ×, direction of axial dipole. Crossing line represents mean axis of magnetization of cleaned specimens (175°, -61°).

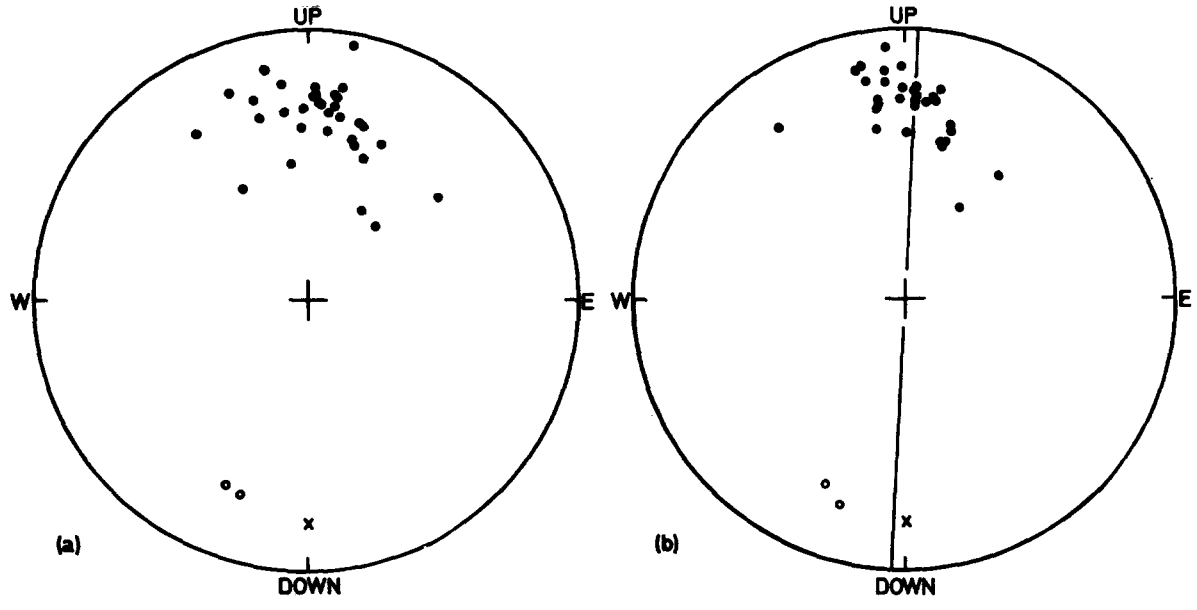


Fig. 2. Directions of magnetization of Scottish baked contacts (a) before cleaning, and (b) after cleaning. Equal area projections. Closed symbols represent southward declinations; open symbols represent northward declinations. ×, direction of axial dipole. Crossing line represents mean axis of magnetization of cleaned specimens (171°, -58°).

Table 5
Mean directions of magnetization and associated statistics. For computational purposes normal dyke N and its contacts have been considered reversed. The VGP derived from all 99 specimens has coordinates 73.4°N, 196.8°E (northern hemisphere) and semi-axes of the oval of confidence [47], $\delta m = 5°$, $\delta p = 4°$.

N = number of samples measured.
D = mean declination of remanent magnetization (ETN).
I = mean inclination of remanent magnetization.
R = length of the resultant of the N unit vectors.
k = estimate of Fisher [37] precision parameter.
α_{95} = semi-angle of cone of 95% confidence for mean direction.
δ = angular standard deviation.

Rocks	N	D (deg)	I (deg)	R	k	α_{95} (deg)	δ (deg)
ac demagnetized dyke specimens	42	171.9	-58.1	39.07	14.0	6.1	21.5
Thermally demagnetized dyke specimens	25	168.9	-59.0	23.16	13.1	8.3	22.1
Thermally demagnetized contact specimens	32	174.6	-60.8	30.86	27.2	5.6	15.3
All of the above specimens	99	172.0	-59.3	93.06	16.5	3.6	20.0

latitude 73.4°N and longitude 196.8°E (northern hemisphere). This is in reasonable agreement with previously obtained British Tertiary pole positions [5, 39-46].

6. PETROLOGY

The opaque petrology of the dyke specimens was examined in thin polished section, initially with a view to discovering statistical correlations between petrology and polarity; but statistical comparison proved impossible since only one of the 21 dykes was normal. Considering opaque minerals only, all dykes contained optically homogeneous titanomagnetite (frequently skeletal) and small quantities of iron sulphide. In addition, two dykes (K and S) contained exsolved titanomagnetite and one (Z) contained discrete hemoilmenite as well as exsolved titanomagnetite. The normal dyke (N) was one of those containing only optically homogeneous titanomagnetite and iron sulphide.

7. CONCLUSION

The magnetic data presented indicate that the reversed magnetizations in these British Tertiary dykes were due to field reversal. The limited amount of petrological data produced no conflicting evidence in favour of self-reversal. The overall system of dykes and baked contacts gave a virtual geomagnetic pole position at 73.4 °N, 196.8°E.

Tertiary geomagnetic field intensities have also been deduced from some of the specimens described here. This work will be reported in a further paper [36].

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